TRIBHUWAN UNIVERSITY

INSTITUTE OF ENGINEERING

PURWANCHAL CAMPUS

A MINOR PROJECT REPORT ON

"GPS-BASED SMART POTHOLE DETECTION, MEMORY, AND ADVANCE DRIVER ALERT SYSTEM"

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DECLARATION

We hereby declare that the project report entitled "GPS-BASED SMART POTHOLE DETECTION, MEMORY, AND ADVANCED DRIVER ALERT SYSTEM", submitted to the Department of Electronics and Computer Engineering, IOE, Purwanchal Campus, in partial fulfillment of the requirements for the degree of Bachelor of Engineering in Electronics, Communication, and Information Technology, is the result of our own work carried out under proper supervision. We confirm that this report has not been submitted, either in part or in full, to any other university or institution for the award of any degree or diploma. Furthermore, we declare that we are the sole authors of this work, and that all sources of information used in this project have been properly acknowledged and referenced.

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ABSTRACT

Road potholes are a significant challenge to transportation systems, leading to vehicle damage, traffic congestion, and increased risk of accidents, especially in developing regions. Traditional pothole detection methods rely on manual inspection and public reporting, which are inefficient, time-consuming, and often inaccurate. To address these issues, this project presents a GPS-based smart pothole detection, memory, and advance driver alert system. The proposed system utilizes an MPU6050 accelerometer and gyroscope sensor to detect abnormal road vibrations caused by potholes using a dynamic thresholding technique based on statistical analysis of acceleration data. A GPS module is integrated to record the precise location of detected potholes. The system stores this information in local memory and optionally in a cloud database, creating a reusable pothole database. An important feature of this system is its ability to provide advance alerts to drivers approximately 200 meters before reaching a known pothole, enhancing driving safety and comfort. The solution is designed to be low-cost, scalable, and suitable for real-world deployment in vehicles such as cars, bikes, and public transport. By combining real-time detection, location-based memory, and predictive alerts, the system contributes to improved road safety, reduced vehicle damage, and smarter infrastructure monitoring.

Keywords: *Pothole detection, Vehicle safety, GPS tracking, MPU6050 sensor*

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LIST OF ABBREVIATIONS

API: Application Programming Interface

GPS: Global Positioning System

IDE: Integrated Development Environment

LED: Light Emitting Diode

ORSM: Open Source Routing Machine

SDK: Software Development Kit

CHAPTER 1

INTRODUCTION

Road potholes are a major cause of vehicle damage, traffic congestion, and road accidents, especially in developing regions where road maintenance is slow and reactive. Traditional road inspection methods depend heavily on manual surveys and public complaints, which are time-consuming, costly, and inefficient.

This project proposes a GPS-based smart pothole detection and memory system that automatically detects potholes while driving, stores their locations, and provides advance alerts to drivers when approaching known potholes on the same road. The system also updates its memory when potholes are repaired, ensuring that alerts remain accurate and relevant over time.

1.1 PROBLEM STATEMENT

Currently, drivers have no reliable way to know about potholes before reaching them. Even when potholes are repaired, outdated information often remains uncorrected. Existing solutions either focus only on detection or require expensive infrastructure.

There is a need for a low-cost, automated system that:

- Detects potholes in real time
- Remembers pothole locations
- Alerts drivers before reaching potholes
- Updates or removes pothole data after repairs

1.2 OBJECTIVES

The main objectives of the project are as follows:

- To design a system that detects potholes using vibration sensors.
- To store detected potholes along with their GPS coordinates.
- To provide advance alerts to drivers.

- To develop a low-cost and scalable solution suitable for real-world deployment.

1.3 Scope of the Project

The proposed system is designed for:

- Cars, bikes, and public transport vehicles.
- Urban and semi-urban road environments.
- Offline operation with optional cloud support.

The project focuses on detection, storage, alerting, and data updating. Actual road repair is outside the scope of this project.

1.4 Limitations

- GPS may be inaccurate in tunnels, urban areas, or under dense tree cover.
- Sensors can misinterpret bumps or speed breakers as potholes.
- Hardware durability may be affected by heat, continuous vibration, and weather conditions.
- Limited memory may restrict the storage of large pothole datasets.
- Poor network connectivity can delay cloud updates or alert synchronization.
- Different vehicles may require calibration for accurate pothole detection.

CHAPTER 2

RELATED THEORY

2.1 HARDWARE

- **ESP32 microcontroller:** ESP32 acts as the main controller of the system. It processes data received from the MPU6050 sensor and GPS module, detects potholes based on sudden vibrations, and manages communication. It can also store data and send alerts or upload information to a server if required.

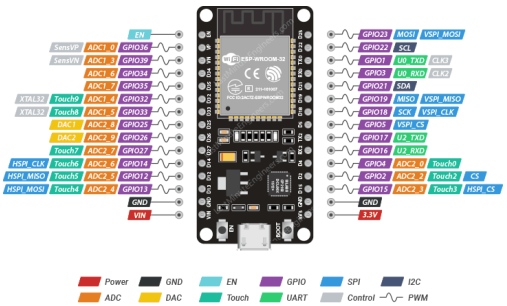

Figure 2.1: ESP 32

- **GPS module (NEO-6M):** The GPS module provides real-time location data (latitude and longitude). When a pothole is detected, the system records its exact location. This helps in creating a database and enables future alerts for drivers approaching the same location.

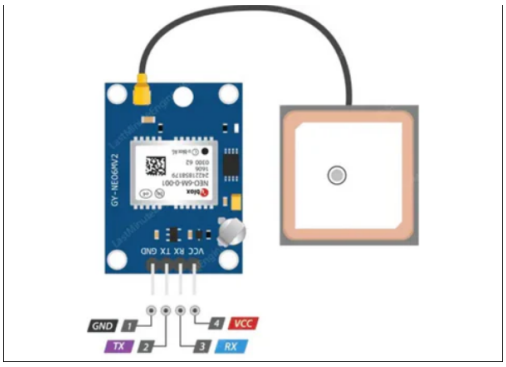

Figure 2.2: GPS module (NEO-6M)

- **MPU6050 accelerometer/gyroscope:** The MPU6050 sensor detects motion and vibration. In this project, it measures sudden changes in acceleration when a

vehicle passes over a pothole. These abnormal vibrations are used to identify the presence of potholes.

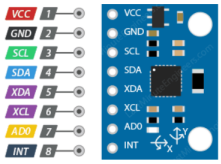

Figure 2.3: MPU6050 accelerometer/gyroscope

- **Power supply**

2.2 SOFTWARE

- **Arduino IDE:** Arduino IDE is an open-source development environment used for writing, compiling, and uploading embedded C/C++ code to the ESP32 microcontroller. It utilizes libraries such as Wire (I2C communication) and TinyGPS++ (GPS data parsing) to interface with hardware modules. In this project, it enables real-time data acquisition from the MPU6050 sensor, processes acceleration thresholds for pothole detection, and manages serial communication with the GPS module and cloud services.
- **Cloud platform (Firebase / local server):** Firebase is a cloud-based Backend-as-a-Service (BaaS) platform that provides real-time database and data synchronization capabilities. In this system, Firebase Realtime Database (or Firestore) is used to store structured data such as GPS coordinates, timestamps, and pothole event logs. It supports REST APIs and SDK integration, enabling seamless data transmission from the ESP32 and real-time retrieval by the mobile application. It also ensures scalability, low latency, and secure data handling.
- **React Native:** React Native is a cross-platform mobile application development framework based on JavaScript and React. It is used to build the user interface (UI) and handle application logic. In this project, it integrates with Firebase using APIs/SDKs to fetch and display pothole data dynamically on maps. It supports features such as real-time notifications, state management, and asynchronous data handling, providing an interactive and responsive user experience.

CHAPTER 3

LITERATURE REVIEW

Road pothole detection has become an important research area due to its impact on road safety, vehicle damage, and traffic efficiency [1]. Traditional pothole identification methods rely on manual inspection and public reporting, which are slow, costly, and often inaccurate [3]. To address these issues, researchers have developed automated pothole detection systems using vibration and acceleration data collected from sensors mounted on vehicles [2]. Accelerometers and gyroscopes are commonly used to detect abnormal vertical movements caused by potholes, though early systems often suffered from false detections due to speed breakers and uneven road surfaces [6].

To improve detection accuracy and usability, many studies have integrated GPS technology with sensor-based systems [2]. GPS-enabled pothole detection allows each pothole to be stored with precise geographic coordinates, enabling road condition mapping and location-based analysis [5]. Some approaches use cloud-based or crowdsourced data to enhance reliability; however, such systems often require continuous internet connectivity and expensive infrastructure [5]. Moreover, most existing solutions focus primarily on detection and mapping rather than assisting drivers in real time [4].

Recent research emphasizes the need for intelligent systems that not only detect potholes but also provide advance alerts and maintain updated pothole records [1]. While a few systems offer driver alerts, they are typically reactive and do not consider pothole repair or data validation over time [2]. Therefore, there remains a need for a low-cost, integrated solution that combines real-time detection, GPS-based memory storage, advance driver alerts, and automatic data updating [1]. The proposed system aims to address these gaps by offering a practical and scalable approach suitable for real-world deployment [2].

CHAPTER 4

METHODOLOGY

4.1 POTHOLE DETECTION

1. Pothole Detection (Hardware / ESP32)

The detection logic in pothole1.ino uses a Dual-Threshold Window Trigger combined with a Simple Moving Average (SMA).

Z-Axis Smoothing (SMA): The raw accelerometer data is smoothed over a window of 10 samples to filter out high-frequency engine noise.

$$Z_{smoothed} = \frac{1}{n}\sum_{i=1}^{n} Z_{relative} \tag{4.1}$$

Detection Algorithm:

Step 1: (The Dip): If $Z_{smoothed} < -Threshold_{Dip}$ (default 150), a potential pothole is "started."

Step 2: (The Impact): If the Z-axis then spikes to $Z_{smoothed} > Threshold_{Impact}$ (default 800) within a 300ms window, a pothole is recorded.

2. Distance Calculation (Haversine Formula)

Used in app/index.tsx within the getDistance function to accurately find the distance between the user and reported potholes over the Earth's curved surface.

Formula:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1)\cdot\cos(\phi_2)\cdot\sin^2\left(\frac{\Delta\lambda}{2}\right) \tag{4.2}$$

$$c = 2\cdot\text{atan2}(\sqrt{a}, \sqrt{1-a}) \tag{4.3}$$

$$d = R\cdot c \quad (\text{where } R = 6{,}371 \text{ km}) \tag{4.4}$$

3. Bearing/Heading Calculation

Used in calculateBearing in app/index.tsx to determine the map's rotation so that the

road ahead is always at the top of the screen (Heading-Up).

Formula:

$$y = \sin(\Delta\lambda) \cdot \cos(\phi_2) \tag{4.5}$$

$$x = \cos(\phi_1) \cdot \sin(\phi_2) - \sin(\phi_1) \cdot \cos(\phi_2) \cdot \cos(\Delta\lambda) \tag{4.6}$$

$$\theta = \text{atan2}(y, x) \tag{4.7}$$

4. Destination Point Calculation

Used in getPointAtDistance in app/index.tsx to project a point a few meters ahead of the user for better camera placement.

Formula:

$$\phi_2 = \text{asin}\left(\sin(\phi_1) \cdot \cos(d/R) + \cos(\phi_1) \cdot \sin(d/R) \cdot \cos(\theta)\right) \tag{4.8}$$

$$\lambda_2 = \lambda_1 + \text{atan2}\left(\sin(\theta) \cdot \sin(d/R) \cdot \cos(\phi_1), \cos(d/R) - \sin(\phi_1) \cdot \sin(\phi_2)\right) \tag{4.9}$$

5. Path Proximity (Route Filtering)

Used in isPointNearPath in app/index.tsx to decide if a pothole is relevant to your current route.

Logic: It iterates through the OSRM path segments and checks if the distance from the pothole to the nearest segment start/end point is less than 15 meters. By doing this, it ignores potholes on parallel side streets.

4.2 GPS-BASED LOCATION TRACKING

A GPS module continuously provides latitude and longitude data. Each detected pothole is stored as a GPS point along with magnitude, time and entry type information.

4.3 POTHOLE MEMORY STORAGE

Pothole data is uploaded to a central database. Each record includes:

- GPS coordinates (latitude and longitude)

- magnitude i.e. the impact value felt by sensor
- reportedAt
- type (hardware or manual entry)

4.4 ADVANCE DRIVER ALERT

While driving, the system calculates the distance between the current vehicle position and stored pothole locations. If an active pothole is detected within 200 meters ahead, an alert is generated using a mobile speaker, screen red flash.

4.5 BLOCK DIAGRAM

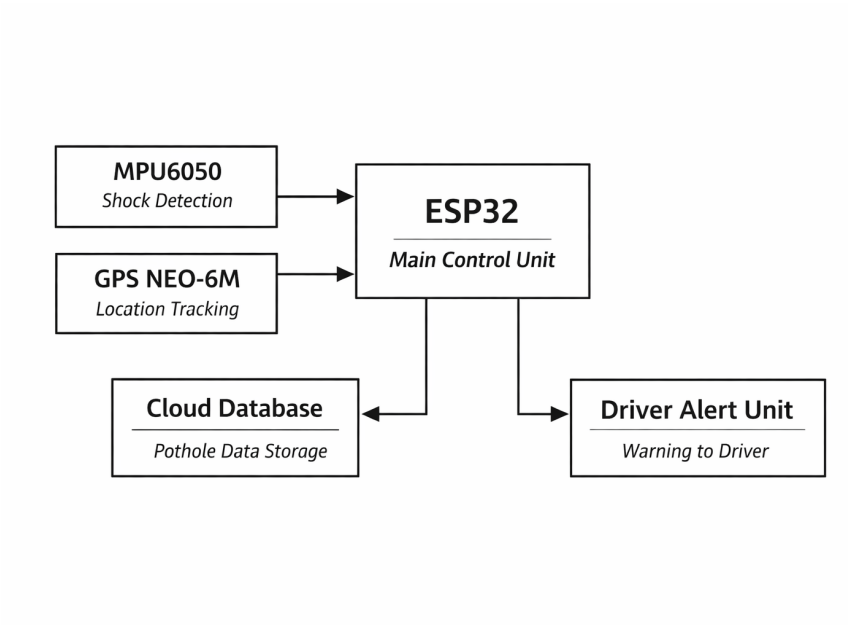

Figure 4.1: Block Diagram

4.6 FLOW CHART

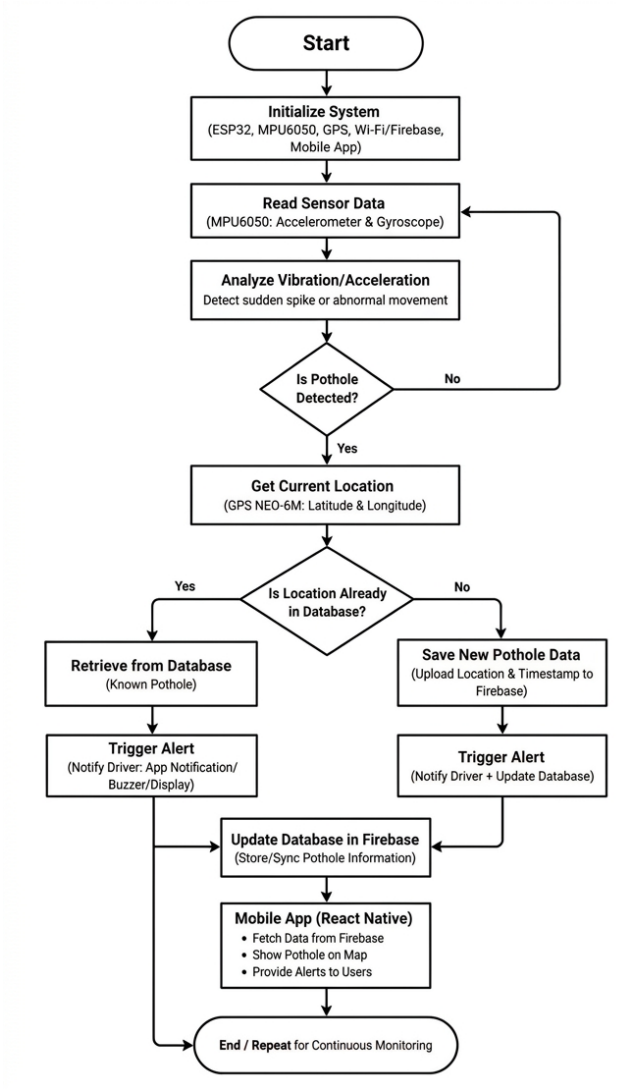

Figure 4.2: Flow Chart

CHAPTER 5

EXPECTED OUTCOMES

The system will deliver an automated and reliable pothole detection and alerting solution by integrating sensors, microcontrollers, communication modules, and data processing technologies. The key outcomes include:

- **Accurate Pothole Detection:** Sensors identify potholes with precision, ensuring that road irregularities are captured consistently under various driving conditions.
- **Timely Alerts for Drivers:** Detected potholes are processed in real time, and drivers receive advance warnings to help them avoid potential damage or unsafe situations.
- **Automatic Data Updates:** Each detected pothole is stored in the database and the app updated the pothole datas automatically every 2 seconds.
- **Improved Safety and Reduced Vehicle Damage:** By alerting drivers early, the system helps minimize accidents, reduce maintenance costs, and contribute to safer driving experiences.

CHAPTER 6

RESULTS

Manual Pothole Entry

The system allows users to manually report potholes by just tapping the map if they are not automatically detected or hard to reach out at the pothole. This feature ensures that all hazardous areas can be recorded in the database, providing additional reliability.

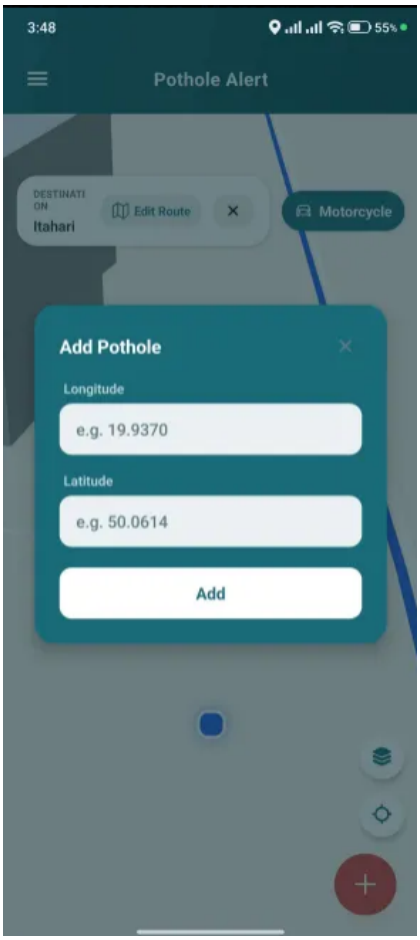

Figure 6.1: Manual Pothole Entry Interface

Admin Login

The administrative panel allows authorized personnel to log in, manage pothole reports, and monitor system performance. It is crucial for maintaining and validating the database of detected potholes.

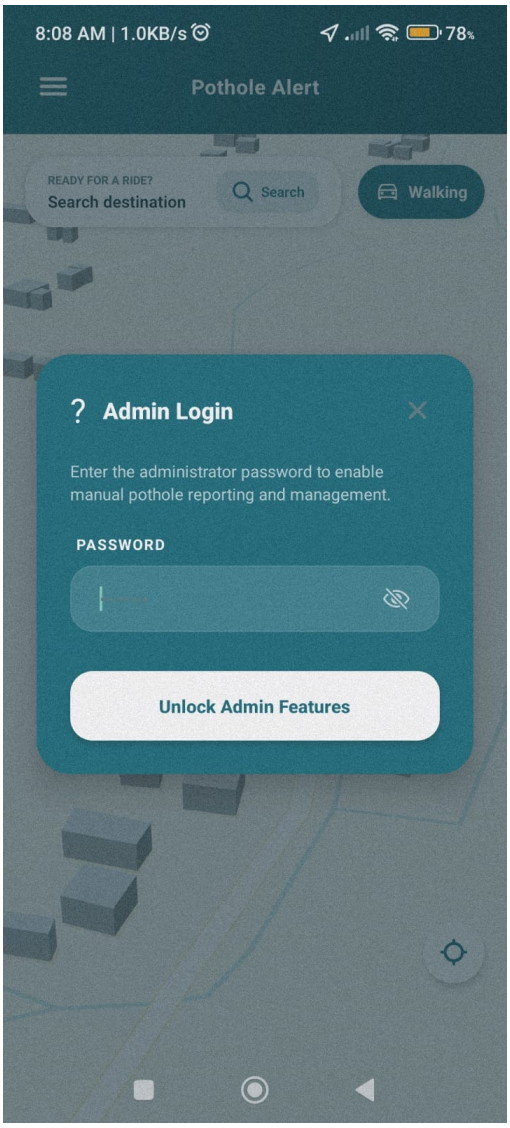

Figure 6.2: Admin Login Interface

Potholes in Map

Detected potholes are displayed on an interactive map with their GPS coordinates. This feature helps drivers anticipate road hazards and provides authorities with a visual overview for maintenance planning.

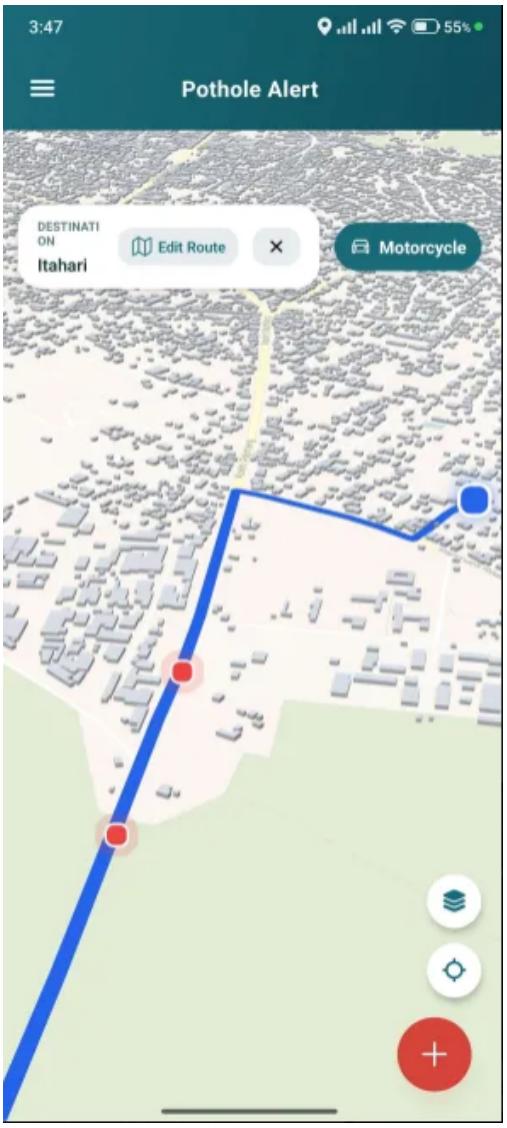

Figure 6.3: Detected Potholes Displayed on Map

Observations

During testing, the system successfully detected and recorded potholes in real time. Manual entry provided a backup for missed detections. The admin panel facilitated efficient monitoring and validation of the pothole database. The GPS mapping feature accurately reflected pothole locations and proved helpful for route planning and infrastructure management.

CHAPTER 7

APPLICATIONS

- **Smart Transportation Systems:** The system helps build safer and smarter transport networks by continuously detecting potholes, storing their GPS locations, and updating their status over time. Vehicles equipped with the system can share road condition data, which supports smoother traffic flow, fewer sudden slowdowns, and better route planning for drivers and fleet operators.
- **Road Maintenance Planning:** The collected pothole data gives authorities an accurate and up-to-date map of road damage. This makes it easier to prioritize repairs, allocate budgets, and schedule maintenance more effectively. Instead of relying on manual surveys or public complaints, agencies can work with reliable and automatically generated data.
- **Driver Assistance Systems:** Drivers receive early warnings when approaching known potholes, reducing the chance of accidents or sudden maneuvers. The alerts also help protect vehicles from damage. This adds an extra layer of safety to existing driver assistance tools and improves the overall driving experience, especially on unfamiliar routes.
- **Smart City Infrastructure:** The system can integrate with smart city platforms to support long-term infrastructure monitoring. Real-time and historical data on road quality can be used for analytics, urban planning, and automated reporting. As more vehicles contribute data, cities gain a scalable, low-cost solution for maintaining safer and better-managed roads.

CHAPTER 8

CONCLUSION AND FUTURE ENHANCEMENTS

Conclusion

The proposed GPS-based smart pothole detection offers an effective solution to enhance road safety and driving comfort. By combining real-time detection with GPS-based location tracking, the system identifies road irregularities, stores their positions, and notifies drivers in advance, reducing accidents and vehicle damage. Unlike manual reporting, the system automates detection and alerts, making it efficient, reliable, and cost-effective. It also provides valuable data for road maintenance and infrastructure planning. Overall, the system addresses driver safety and infrastructure monitoring challenges and has the potential for large-scale implementation in smarter transportation networks.

Future Enhancements

- Integration of AI/ML algorithms for more accurate pothole detection and classification.
- Use of computer vision (camera-based detection) to improve reliability.
- Energy optimization using low-power hardware or solar support.
- Implementation of severity analysis to prioritize dangerous potholes.
- Cloud-based data analytics for city-wide pothole mapping and predictive maintenance.

REFERENCES

[1] K. V. R. Reddy, M. S. Kumar, B. Vamshi, and J. Balaji, "IoT-based pothole detection system," *International Journal of Engineering Research and Science & Technology*, vol. 21, no. 3, pp. 724–730, 2025.

[2] D. B. Bhoyar, S. K. Mohod, N. Ambatkar, P. Selokar, and A. R. Bhagat Patil, "A comprehensive system for pothole detection and driver alert," *International Journal of Basic and Applied Sciences*, 2025.

[3] D. A. Casas-Avellaneda and J. F. López-Parra, "Detection and localization of potholes in roadways using smartphones," *Dyna*, vol. 83, no. 195, pp. 156–162, 2016.

[4] R. Nienaber, M. Booysen, and R. Kroon, "Detecting potholes using simple image processing techniques and real-time data," in *Proc. IEEE Intelligent Transportation Systems Conference (ITSC)*, 2015, pp. 1–6.

[5] H. Maeda, Y. Sekimoto, T. Seto, T. Kashiyama, and H. Omata, "Road damage detection using deep neural networks with images captured through a smartphone," in *Proc. IEEE International Conference on Big Data (Big Data)*, 2018, pp. 511–520.

[6] S. Silva, R. S. N. de Silva, and K. A. S. Fernando, "Pothole detection using machine learning algorithms," in *Proc. International Conference on Advances in ICT for Emerging Regions (ICTer)*, 2017, pp. 1–6.

[7] Y.-M. Kim, Y.-G. Kim, S.-Y. Son, S.-Y. Lim, B.-Y. Choi, and D.-H. Choi, "Review of recent automated pothole-detection methods," *Applied Sciences*, vol. 12, no. 11, p. 5320, 2022.

[8] C. Wu, Z. Wang, S. Hu, J. Lepine, X. Na, D. Ainalis, and M. Stettler, "An automated machine-learning approach for road pothole detection using smartphone sensor data," *Sensors*, vol. 20, no. 19, p. 5564, 2020.

[9] S. Zanevych, V. Yovvak, O. Basystiuk, N. Shakhovska, S. Fedushko, and S. Argyroudis, "Evaluation of pothole detection performance using deep learning models under low-light conditions," *Sustainability*, vol. 16, no. 24, p. 10964, 2024.

[10] A. Nataraju, A. C. J., L. Arun Kumar, H. B. Sanjay, and H. H. V., "Automatic pothole and humps detection system of roads using CNN and GPS," *Journal of Firewall Software and Networking*, 2023.